Yuxi Mi 糜予曦

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Education and Appointments

Fudan University **2020 – Present**
Postdoctoral Researcher, College of Computer Science and Artificial Intelligence (mentor: Prof. Shuigeng Zhou). 2024 – Present
Ph.D. in Computer Science, School of Computer Science (advisor: Prof. Shuigeng Zhou). 2020 – 2024

Institute of Information Engineering, Chinese Academy of Sciences **2017 – 2020**
M.Sc. in Cybersecurity, State Key Laboratory of Information Security (advisor: Prof. Wentao Zhang).

Shandong University **2013 – 2017**
B.Sc. in Information Security, School of Mathematics.

Research Experience

Ph.D. Student → Postdoc, Shanghai Key Laboratory of Intelligent Information Processing **2020 – Present**

- Conducted doctoral research on privacy-preserving face recognition under Prof. Shuigeng Zhou, in collaboration with Tencent Youtu Lab. Research interest includes face image protection, biometric template protection, and synthetic face recognition, with publications at CVPR, ICCV, AAAI, ACM MM, ICLR, and in Information Fusion. Ph.D. dissertation: Research on Privacy-Preserving Deep Face Recognition Algorithms.
- Continued face and iris recognition research as a postdoc, with publications at CVPR and AAAI. Received the NSFC Young Scientists Fund (Cat. C), the Postdoctoral Fellowship Program of CPSF (Cat. C), and the Fudan Super Postdoctoral Program. Taught the graduate course “Principles and Applications of Generative Models”.

M.Sc. Student, State Key Laboratory of Information Security, IIE, CAS **2017 – 2020**

- Studied symmetric cryptographic, with a focus on provable security of authenticated encryption algorithms and security analysis of NIST Lightweight Cryptography (LWC) competition candidate algorithms. Master's thesis: Research on Sponge-Based Lightweight Authenticated Encryption Algorithms.

Project Experience

Biometric Algorithms, Tencent Youtu Lab **2022 – Present**

- Research intern (2022–2024); project lead (2024–present). Having **9 publications at CVPR, ICCV, AAAI, ACM MM, ICLR, and in Information Fusion** (7 as first / co-first), 2 submissions, and **5 patent applications** (1 granted).
- Designed face protection methods that resist reconstruction attacks and maintain SOTA recognition performance, based on frequency decomposition, trainable residuals, spherical interpolation, and diffusion noise.
- Developed three approaches to address controllability, diversity, and fairness of synthetic face generation. The synthetic data perform within <1% of the real-data baseline; related work won the CVPR 2024 FRCSyn Challenge.
- Introduced the first VR open scenes iris dataset & benchmark, and a normalization-free recognition paradigm; nominated as **CVPR 2026 best paper award candidate** (0.4% of submissions) and oral presentation (0.79%).

Federated Learning for Financial Applications, Tencent Weixin Group **2021 – 2023**

- Research intern focused on privacy and robustness in federated learning. Having **4 publications** in IEEE T-AI and Neural Networks and at APWeb-WAIM (3 as first author), and **1 patent application**.
- Introduced the first differentially private vertical federated learning framework supporting multiple data modalities.
- Designed a defense against backdoor attacks based on feature-map anomaly detection, and developed a semi-supervised approach that incorporates server-side knowledge and low-confidence client samples.

Ship Recognition from Sonar Signals, CSSC No. 715 Research Institute **2025**

- Project lead. Deployed angular-margin time-frequency learning for ship identification from passive sonar signals. Achieved >90% identification accuracy and >94% known/unknown ship discrimination accuracy.

Publications *=co-first author ‡=corresponding author

1. **Y Mi***, Q Yuan*, Z Zhong, X Zhao, J Zhou‡, F Zhu‡, J Guan, S Zhou‡. ImmerIris: A Large-Scale Dataset and Benchmark for Off-Axis and Unconstrained Iris Recognition in Immersive Applications. **CVPR 2026 Award Candidate (0.4%) & Oral (0.79%)**. (CCF-A)
2. **Y Mi**, Q Yuan, J Xu, Y Zhou, X Zhao, J Wang, R Guo, S Zhou‡. SteerFace: Debiasing Synthetic Face Generation via Adaptive Residue Perturbation. **ACM MM 2026**. (CCF-A)

3. **Y Mi**, Z Zhong, Y Huang‡, Q Yuan, X Zhao, J Xu, S Ding, S Wang, R Guo, S Zhou‡. Data Synthesis with Diverse Styles for Face Recognition via 3DMM-Guided Diffusion. **CVPR 2025**. (CCF-A)
4. **Y Mi***, Z Zhong*, Y Huang‡, J Ji, J Xu, J Wang, S Wang, S Ding, S Zhou‡. Privacy-Preserving Face Recognition Using Trainable Feature Subtraction. **CVPR 2024**. (CCF-A)
5. **Y Mi***, Y Huang*, J Ji, M Zhao, J Wu, X Xu, S Ding, S Zhou‡. Privacy-Preserving Face Recognition Using Random Frequency Components. **ICCV 2023**. (CCF-A)
6. **Y Mi**, Y Huang, J Ji, H Liu, X Xu, S Ding, S Zhou‡. DuetFace: Collaborative Privacy-Preserving Face Recognition via Channel Splitting in the Frequency Domain. **ACM MM 2022**. (CCF-A)
7. Z Zhong*, **Y Mi***, Y Huang‡, J Xu, G Mu, S Ding, J Zhang, R Guo, Y Wu, S Zhou‡. SlerpFace: Spherical Linear Interpolation for Face Template Protection. **AAAI 2025**. (CCF-A)
8. **Y Mi**, H Liu, Y Xia, Y Sun, Y Tang, J Guan, S Zhou‡. FedFlex: Protecting Shared Features in Vertical Federated Learning via Differential Privacy. **IEEE Transactions of Artificial Intelligence 2025**. (IF=7.3)
9. **Y Mi**, Y Sun, J Guan, S Zhou‡. Identifying Backdoor Attacks in Federated Learning via Anomaly Detection. **APWeb-WAIM 2023**. (CCF-C)
10. **Y Mi**, Y Mu, S Zhou‡, J Guan. FedMDR: Federated Model Distillation with Robust Aggregation. **APWeb-WAIM 2021**. (CCF-C)
11. H Liu, **Y Mi**, Y Tang, J Guan, S Zhou‡. Boosting Semi-Supervised Federated Learning by Effectively Exploiting Server-Side Knowledge and Client-Side Unconfident Samples. **Neural Networks 2025**. (CCF-B, JCR Q1, IF=6.3)
12. X Zhao*, Z Zhang*, Y Huang, **Y Mi**, G Mu, S Ding‡, J Wang, R Guo, S Zhou‡. GloTok: Global Perspective Tokenizer for Image Reconstruction and Generation. **AAAI 2026 Oral (4.5%)**. (CCF-A)
13. X Lin*, Y Huang*, J Xu, **Y Mi**, S Zhou, S Ding‡. UIFace: Unleashing Inherent Model Capabilities to Enhance Intra-Class Diversity in Synthetic Face Recognition. **ICLR 2025**. (CCF-A)
14. M Zhao, Y Liu, W He, B Yu, **Y Mi**, S Zhou‡. Towards High Robust Vision-Language Large Models: Benchmark and Method. **ACM MM 2025**. (CCF-A)
15. Ivan DeAndres-Tame‡, Ruben Tolosana, Pietro Melzi, ..., **Y Mi**, et al., Second FRCSyn-Ongoing: Winning Solutions and Post-Challenge Analysis to Improve Face Recognition with Synthetic Data. **Information Fusion 2025**. (JCR Q1, IF=17.9)
16. **Y Mi***, Z Zhong*, Y Huang, S Ding, J Guan, S Zhou‡. Protecting Face Biometric Templates Against Semantic Reconstruction Attack. IJCV Submission.
17. Y Zhou*, **Y Mi***, J Xu, Q Yuan, X Zhao, S Ding, R Guo, J Wang, J Zhang, S Zhou‡. InvFace: Privacy-Preserving Face Recognition via Identity-Distilling Inversion. ICLR Submission.
18. X Zhao, Z Zhang, **Y Mi**, Y Zhou, J Yan, G Mu, S Ding, J Wang, S Zhou. SDA-Codec: Encoder-Side Semantic Differentiation via Alignment for Ultra-Low-Bitrate Image Compression. AAAI Submission.

Patents

1. Y Huang, Z Zhong, **Y Mi**, et al. Image feature processing method, apparatus, and storage medium. Grant No. ZL202410041219.X.
2. **Y Mi**, S Zhou, Y Tang, et al. Anomaly processing method, apparatus and electronic device. Application No. CN202311237548.3.
3. **Y Mi**, Y Huang, J Ji, et al. Identity recognition method, image processing method, and device. Application No. CN202211031084.6.
4. Y Huang, **Y Mi**, J Xu, et al. Face recognition method, apparatus, and storage medium. Application No. CN202410096230.6.
5. Y Huang, **Y Mi**, J Ji, et al. Image processing method, apparatus, and computer program product. Application No. CN202211521684.0.
6. Y Huang, Q Yuan, **Y Mi**, et al. Iris recognition method, apparatus, and computer program product. Application No. CN202610003728.2.

Grants, Funding and Awards

Grants **NSFC Young Scientists Fund (Category C) | PI | 2026 – 2028**
Privacy-Preserving Face Recognition via Semantically Guided Face Synthesis. Grant No. 62506080.

Funding **CPSF Postdoctoral Fellowship Program (Category C) | 2024 – 2026**
Research on Privacy-Preserving Training Data Generation for Face Recognition. Grant No. GZC20251072.
Fudan University Super Postdoctoral Program

Awards **Champion**, CVPR 2024 Face Recognition Contest on Synthetic Data (FRCSyn).
Outstanding Graduate Scholarship, Fudan University: two First-Class and two Second-Class awards (2020-2024).

Academic Activities

Instructor, Principles and Applications of Generative Models (Fall 2025, Fudan University).

TA, Computational Geometry Algorithms (Summer 2021)**,** Web Software Development Technologies (Fall 2021, Fudan University).

Committee Member, Technical Committee on Biometric Recognition, China Society of Image and Graphics (CSIG-BR).

Conference Reviewer, NeurIPS (**Top Reviewer 2025**, 10.2%), CVPR, ICCV, ECCV, ICLR, AAAI, ACM MM, IJCAI.

Journal Reviewer, IEEE T-PAMI, T-IFS, T-DSC, T-BIOM, T-MI, T-NNLS, T-AI.